



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru - 560064



AI-BASED TOOL FOR PRELIMINARY DIAGNOSIS OF DERMATOLOGICAL MANIFESTATIONS

A PROJECT REPORT

Submitted by

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Under the guidance of,

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IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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BONAFIDE CERTIFICATE

Certified that this report "AI-Based Tool For Preliminary Diagnosis Of Dermatological Manifestations" is a bonafide work of LAKSHMI PRIYA NARAYAN(20221CSE0374), HEMANTH K(20221CSE0046) and VANKAM VISWA SAI(20221CSE0045) who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.


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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named Lakshmi Priya Narayan, Hemanth K and Vankam Viswa Sai, hereby declare that the project work titled "AI-Based Tool For Preliminary Diagnosis Of Dermatological Manifestations" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

Skin diseases are a universal health issue worldwide where detecting cases early is required yet traditional methods are often slow, subjective and resource constrained. This project is a deep learning-based system that classifies dermatological lesions by analyzing images. The entire mechanism operates using a custom CNN model that has been modified and trained on TensorFlow and Keras until it starts acting. The model performs training and testing until the best accuracy is achieved and saves it. The initial data was highly unbalanced (by a large margin benign over melanoma), hence RandomOverSampler was introduced to replicate the infrequent samples to balance the classes. The dataset used consists of images that are 28x28 grey images. The best model is saved as best_model.h5 including a clean, multi-page Streamlit web app which depicts the model's performance in the best way possible.

The user is provided with an opportunity to upload a picture of a skin lesion in the web app, and the results will appear on the screen consuming least amount of time. It performs accurately with an accuracy of 97.79%, it downsizes the image to 28x28 and normalizes the pixel values directly as it would be during training, and sends it directly to the model. The results appear within a few seconds with the classification (benign or malignant) and a definite level of confidence and a brief and responsible note that users should seek the advice of a dermatologist about anything. Such performance on a task as important as melanoma detection, causes the entire system to feel sound and prepared to be used in the actual world. This is justified with a sophisticated classification report and Macro-average Area Under the Curve (AUC) of 1.00, which is close to perfect value with all the seven disease groups. The model presents the feasibility of a compact, AI-based device that can be used as an affordable, scalable, and reliable auxiliary diagnostic tool, which can contribute immensely to computer-aided diagnosis and teledermatology creation.